## 6.3.3 Insert

As mentioned, cases #2 and #3 are the most complicated to handle, especially for method `insert`. When we find a partial match between a key and an edge, we will need to break the edge's label down, split the edge into 2 new edges, and add a new node in the middle, in correspondence to the longest common prefix sP.

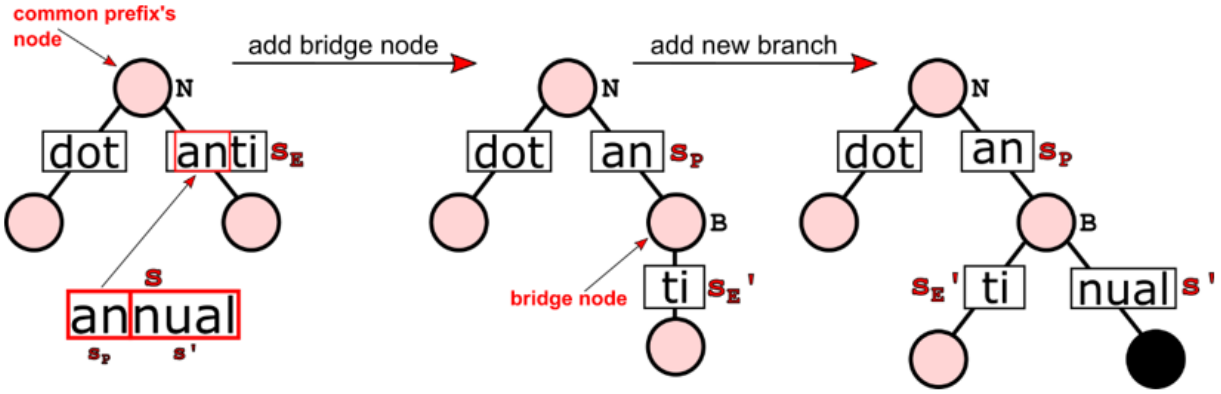



**Figure 6.17 Handling case #2 in edge matching while performing an insertion. In this example, we add we add the word "annual" to a node containing an edge labeled with "anti". To do so, we insert a bridge node B, linked to N by an edge labeled with the common prefix "an", and then two new edges leaving B.**

This is illustrated in figure 6.17: once the common prefix has been found, we need to add a new node in order to split the edge partially matching the string to insert, and then we can add a new branch to this new node.

**Listing 6.15 method insert for Radix Tries**